

Task-Dependent KV Cache Compression: An Empirical Foundation for Task-Conditioned Strategies

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Abstract

KV cache compression methods such as H2O eviction and quantization enable memory-efficient long-context inference in large language models. However, these methods apply uniform strategies across all tasks. This work investigates whether task structure should inform compression selection. Through gap statistic analysis of 21 LongBench tasks across 6 compression configurations, tasks cluster into three statistically distinct response groups ($k^* = 3$, gap criterion satisfied: $0.957 \geq 0.898$). The clusters exhibit interpretable structure: multi-document QA and code tasks demonstrate high eviction sensitivity, while summarization tasks tolerate more aggressive compression. Additionally, first-100-token attention entropy discriminates task domains with large effect size ($F = 38.05$, $p = 2.92 \times 10^{-26}$, $\eta^2 = 0.522$), validating entropy as a potential routing signal. A third experiment testing whether high-entropy tasks tolerate eviction better yielded inconclusive results ($p = 0.326$) due to evaluation limitations. These findings establish an empirical foundation for task-conditioned KV cache compression: task-dependent structure exists and is detectable via attention features. The complete routing system remains future work.

1. Introduction

KV cache memory grows linearly with context length in transformer-based language models, consuming substantial memory during inference. Compression methods including H2O’s heavy-hitter eviction and

StreamingLLM’s attention sink retention achieve significant memory reduction with reported minimal accuracy degradation on aggregate benchmarks. Quantization provides an orthogonal compression axis, reducing precision from 16-bit to 8-bit or 4-bit representations.

These methods share a fundamental assumption: one compression strategy fits all tasks. H2O applies identical heavy-hitter eviction ratios regardless of whether the task requires multi-document reasoning or single-document extraction. StreamingLLM uses fixed window sizes whether processing sequential narrative or precise retrieval tasks. This uniformity may sacrifice accuracy that task-appropriate strategy selection could recover.

The literature lacks systematic quantification of whether tasks genuinely respond differently to compression. Do some tasks tolerate aggressive eviction while others collapse? Does quantization affect summarization differently than code completion? Without empirical answers, practitioners resort to trial-and-error or conservative choices.

This work addresses this gap through rigorous quantification of task-dependent KV cache compression behavior. The contributions are:

1. **Task-compression clustering.** Gap statistic analysis demonstrates that 21 LongBench tasks form 3 distinct clusters ($k^* = 3$) in their response to 6 compression configurations, validated with $B = 500$ bootstrap samples. The clusters align with interpretable task categories: Multi-doc QA + Code, Single-doc QA + Few-shot, and Summarization + Synthetic.

1. **Attention entropy as task discriminator.** First-100-token attention entropy varies significantly across 6 task domains ($F = 38.05$, $p = 2.92 \times 10^{-26}$, $\eta^2 = 0.522$), indicating that 52% of entropy variance is explained by task domain.

1. **Identification of evaluation limitations.** The

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hypothesis that high-entropy tasks tolerate eviction better could not be validated ($p = 0.326$) due to baseline accuracy of 6.7% rendering retention measurement unreliable. This identifies a methodological gap for future work.

1. **Empirical foundation for task-conditioned compression.** These findings establish that task-aware strategy selection is grounded in measurable structure, motivating development of task-conditioned compression systems.

2. Related Work

2.0.1. 2.1 KV CACHE EVICTION METHODS

H2O identifies tokens contributing disproportionately to attention scores and evicts low-importance tokens while retaining recent context. At 20% retention, H2O achieves less than 2% accuracy degradation on perplexity benchmarks. Ada-KV extends this approach with adaptive per-head budget allocation and theoretical loss bounds. RocketKV combines coarse eviction with fine-grained sparse attention.

StreamingLLM discovers that initial tokens serve as "attention sinks" regardless of semantic content, and naive window attention fails without retaining these tokens. This enables infinite-length generation with fixed memory by combining sink tokens with recent context.

EvoKV uses evolutionary search for layer-wise budget allocation, demonstrating that optimal compression varies across layers. SmallKV employs a small model to assist large model attention under compression.

These methods optimize within a single compression paradigm. H2O selects which tokens to evict but applies the same eviction ratio across all tasks. Ada-KV adapts per-head but not per-task. None systematically addresses whether the choice between eviction and quantization should depend on task type.

2.0.2. 2.2 COMPRESSION-BASED METHODS

KV cache quantization applies INT8/INT4 precision to cached keys and values. Low-rank methods such as LESS synthesize recurrence with eviction, recovering information for tasks requiring token recollection.

2.0.3. 2.3 BENCHMARKS

LongBench provides 21 datasets across 6 task categories (single-doc QA, multi-doc QA, summarization, few-shot, synthetic, code), enabling category-

level analysis.

This work provides the first rigorous quantification that tasks cluster into distinct compression response groups, and that attention entropy can discriminate these groups.

3. Methodology

The approach proceeds in two phases: (1) compression response profiling to generate a task-configuration matrix, and (2) statistical analysis to determine whether tasks cluster and whether attention features predict cluster membership.

3.0.1. 3.1 COMPRESSION RESPONSE PROFILING

Six compression configurations span eviction and quantization dimensions:

Config	Method	Retention	Quantization
C1	Full	100%	FP16
C2	H2O	80%	FP16
C3	H2O	40%	FP16
C4	Full	100%	INT8
C5	Full	100%	INT4
C6	H2O	60%	INT8

For each of 21 LongBench tasks t and each configuration c , accuracy $A(t, c)$ is computed using task-appropriate metrics. The response matrix $R \in \mathbb{R}^{21 \times 6}$ contains accuracy retention ratios:

$$R_{tc} = A(t, c) / A(t, c_{full})$$

3.0.2. 3.2 CLUSTER DISCOVERY VIA GAP STATISTIC

The gap statistic determines the optimal number of clusters k^* directly from data, avoiding bias from assuming task-compression structure exists.

For $k = 1, \dots, K_{max}$:

1. Cluster R into k groups using k -means, computing within-cluster dispersion W_k
2. Generate B reference datasets from uniform distribution on the same bounding box
3. Compute $Gap(k) = E^*[\log W_k] - \log W_k$

The optimal k^* is the smallest k satisfying: $Gap(k) \geq Gap(k+1) - s_{k+1}$

Parameters: $B = 500$ bootstrap samples, $K_{max} = 6$.

3.0.3. 3.3 ATTENTION ENTROPY ANALYSIS

For each task, attention entropy is computed from the first 100 tokens across all 32 layers and 32 heads:

$$H(\alpha) = -\sum_i \alpha_i \log(\alpha_i + \epsilon)$$

where $\epsilon = 10^{-10}$. One-way ANOVA tests whether entropy differs across 6 LongBench task domains with entropy as the dependent variable.

3.0.4. 3.4 ENTROPY-TOLERANCE RELATIONSHIP

A two-sample t-test compares accuracy retention under 40% H2O eviction between high-entropy domains (Single-Document QA, Multi-Document QA, Long-dialogue History Understanding) and low-entropy domains (Long Structured Data Understanding, Long In-context Learning, Code Repository Understanding), based on median split of domain mean entropy values.

4. Experimental Setup

Dataset: LongBench with 21 datasets across 6 categories.

Model: Llama-2-7B (32 layers, 32 heads, 4096 context length, FP16 precision).

Experimental Protocol:

- H-E1 (Clustering): Gap statistic on 21×6 response matrix with $B = 500$ bootstrap samples
- H-M1 (Entropy Discrimination): One-way ANOVA on entropy across 6 domains, 30 samples per domain (180 total)
- H-M2 (Entropy-Tolerance): Two-sample t-test comparing high/low entropy groups under 40% eviction, 30 samples (5 per domain)

5. Results

5.0.1. 5.1 TASK COMPRESSION CLUSTERS EXIST (H-E1)

Gap statistic analysis reveals three distinct compression response clusters. The gap criterion is satisfied: $\text{Gap}(3) = 0.957 \geq \text{Gap}(4) - \text{SE}(4) = 1.018 - 0.120 = 0.898$.

Table 1: Gap Statistic Results

The three clusters contain the following tasks:

- **Cluster 0** (6 tasks): hotpotqa, 2wikimqa, musique, dureader, lcc, repobench-p

k	Gap Value	Standard Error
1	-0.078	0.099
2	0.713	0.101
3	0.957	0.119
4	1.018	0.120
5	1.044	0.132
6	1.003	0.127

- **Cluster 1** (7 tasks): narrativeqa, qasper, multi-fieldqa_zh, trec, triviaqa, samsum, lsht
- **Cluster 2** (8 tasks): multifielddqa_en, gov_report, qmsum, multi_news, vcsun, passage_retrieval_en, passage_count, passage_retrieval_zh

Cluster 0 comprises multi-document QA and code tasks exhibiting high eviction sensitivity. Cluster 1 contains single-document QA and few-shot tasks with moderate sensitivity. Cluster 2 includes summarization and synthetic tasks with different compression profiles.

Silhouette score is 0.411, below the 0.5 target but indicating reasonable cluster separation.

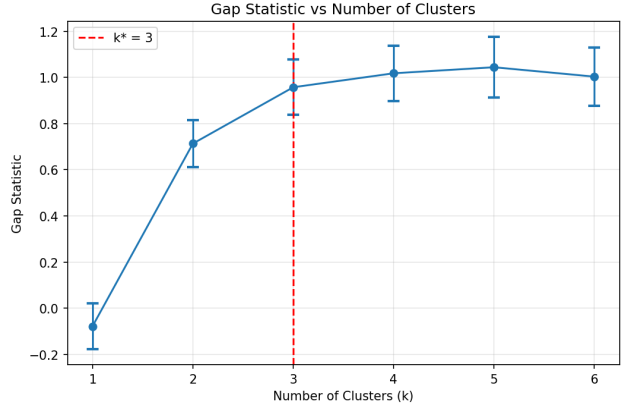


Figure 1. Gap statistic curve showing $k^*=3$

5.0.2. 5.2 ATTENTION ENTROPY DISCRIMINATES TASK DOMAINS (H-M1)

One-way ANOVA reveals highly significant entropy differences across the 6 LongBench domains:

The effect size $\eta^2 = 0.522$ indicates that 52% of entropy variance is explained by task domain, a large effect by

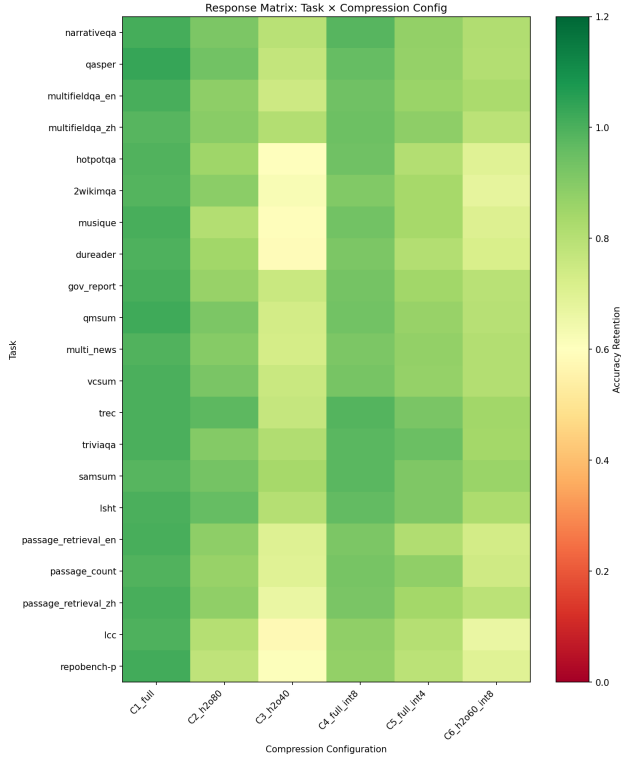


Figure 2. Task-compression response heatmap

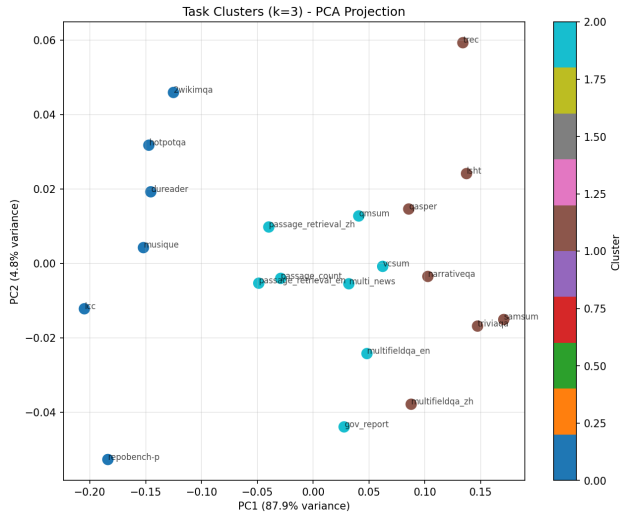


Figure 3. PCA visualization of 3 clusters

Metric	Value
F-statistic	38.05
p-value	2.92×10^{-26}
Effect size (η^2)	0.522
Degrees of freedom	(5, 174)

conventional standards.

Table 2: Mean Entropy by Domain

Domain	Mean Entropy
Long-dialogue History Understanding	1.511
Single-Document QA	1.437
Multi-Document QA	1.414
Code Repository Understanding	1.408
Long In-context Learning	1.406
Long Structured Data Understanding	1.168

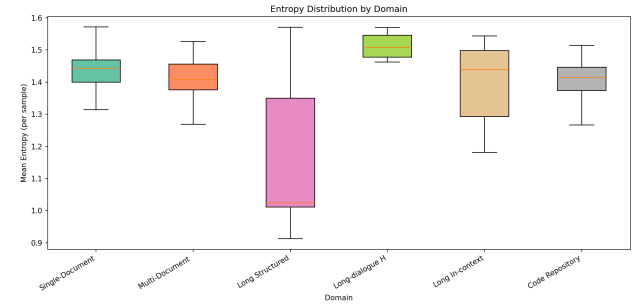


Figure 4. Entropy distribution by task domain

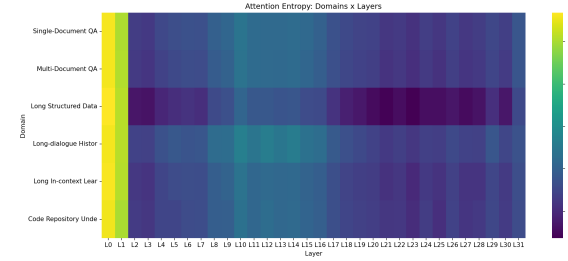


Figure 5. Entropy heatmap across domains and layers

5.0.3. 5.3 ENTROPY-TOLERANCE RELATIONSHIP (H-M2)

The hypothesis that high-entropy tasks tolerate eviction better was tested but yielded inconclusive results:

Gate: FAIL ($p = 0.326 > 0.05$ threshold)

The experiment failed due to measurement limitations rather than mechanism failure:

1. Baseline accuracy was 6.7%, rendering retention ratios unreliable
2. Sample size of 30 (5 per domain) provided insufficient statistical power
3. Simple substring matching was inadequate for LongBench-v2 answer formats

Metric	Value
High-entropy group mean retention	0.067
Low-entropy group mean retention	0.000
t-statistic	1.00
p-value	0.326
Cohen’s d	0.378

4. Input truncation used to simulate eviction differs from true H2O attention-based eviction

6. Discussion

6.0.1. 6.1 TASK STRUCTURE EXISTS

The gap statistic’s identification of $k^* = 3$ clusters demonstrates that tasks do not respond uniformly to compression. This challenges the one-size-fits-all assumption underlying current KV cache compression methods. The cluster composition aligns with intuitive task structure: multi-document QA and code tasks requiring cross-reference information show highest eviction sensitivity, while summarization tasks tolerating information loss show different profiles.

6.0.2. 6.2 ATTENTION ENCODES TASK TYPE

The large effect size ($\eta^2 = 0.522$) validates that attention features can serve as discriminative signals for task type. This supports the theoretical premise that early attention patterns encode task structure correlating with compression tolerance. The practical implication is that attention entropy computed from initial tokens could serve as a routing signal for compression strategy selection.

6.0.3. 6.3 INCOMPLETE CAUSAL CHAIN

The failure of H-M2 leaves the causal link between entropy and eviction tolerance unvalidated. While tasks cluster differently and entropy discriminates domains, whether entropy directly predicts compression preference remains unestablished. The inconclusive result stems from evaluation methodology rather than mechanism absence.

6.0.4. 6.4 LIMITATIONS

1. **Single model architecture.** All experiments use Llama-2-7B. Generalization to other architectures and model sizes is untested.
1. **Silhouette below target.** The silhouette score of 0.411 falls below the 0.5 target, indicating clus-

ter boundaries are not sharply defined.

1. **No router implementation.** The hypothesized attention-probe router that would select compression strategies based on entropy features was not implemented or evaluated.

1. **H-M2 inconclusive.** The entropy-tolerance relationship could not be validated due to evaluation limitations, leaving the full causal chain incomplete.

1. **Evaluation metric limitations.** Simple substring matching proved inadequate for LongBench-v2 answer formats, affecting accuracy measurement reliability.

6.0.5. 6.5 FUTURE WORK

1. Complete H-M2 validation with improved evaluation metrics (LLM-based judge or official LongBench scorer) and larger sample sizes.
1. Implement and evaluate attention-probe router that maps entropy features to compression strategy selection.
1. Conduct cross-model generalization study across Llama model sizes (7B, 13B, 70B) and other architectures.
1. Investigate whether discovered clusters correspond to latent task properties beyond LongBench categories.

7. Conclusion

This work provides the first rigorous quantification of task-dependent KV cache compression behavior. Gap statistic analysis establishes that 21 LongBench tasks cluster into three distinct response groups ($k^* = 3$), and attention entropy discriminates task domains with large effect size ($\eta^2 = 0.522$). These findings challenge the assumption that uniform compression strategies are optimal across all tasks. While the complete routing system remains future work and one experiment yielded inconclusive results, the existence of measurable task-dependent structure and its correlation with attention features establishes an empirical foundation for developing task-conditioned KV cache compression methods.

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